

County-Level Economic Trends Analysis (2017–2022)

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Abstract

This report analyzes county-level economic trends in the United States from 2017 to 2022, visualized through a Tableau dashboard. Using data collected from the U.S. Census Bureau's American Community Survey (ACS), the analysis explores percentage changes in population, employment, and median house prices across Metropolitan, Micropolitan, and Rural areas. The objective is to identify patterns of economic growth and spatial disparities, providing insights into regional development over time.

1 Introduction

Regional economic trends offer valuable insight into population dynamics and development patterns across the United States. This study uses county-level data to compare how urban and rural areas evolved economically between 2017 and 2022. By analyzing changes in key indicators—population, employment, and housing price—the report highlights where growth has concentrated and how structural differences influence regional performance.

2 Data and Methodology

2.1 Data Sources

The dataset was derived from the U.S. Census Bureau's American Community Survey (ACS) using the `tidycensus` package in R. Three main indicators were extracted for all U.S. counties:

- **Population (2017–2022)** – ACS variable B01003_001
- **Employment (2017–2022)** – ACS variable B23025_001
- **Median House Price (2017–2022)** – ACS variable B25077_001

A classification dataset was joined to assign each county to a Metropolitan, Micropolitan, or Rural category.

2.2 Data Processing

The following transformations were performed in R:

- Merged datasets by `GEOID` (county identifier).

- Calculated percentage changes between 2017 and 2022 for all indicators:

$$Change\ Rate = \frac{Value_{2022} - Value_{2017}}{Value_{2017}} \times 100$$

- Replaced missing classifications with “Rural” using `replace_na()`.
- Filtered out invalid or non-contiguous U.S. counties.

3 Dashboard Overview

The Tableau dashboard visualizes economic change across the U.S. at the county level. The interactive visuals allow comparison by area type, indicator, and geographic distribution.

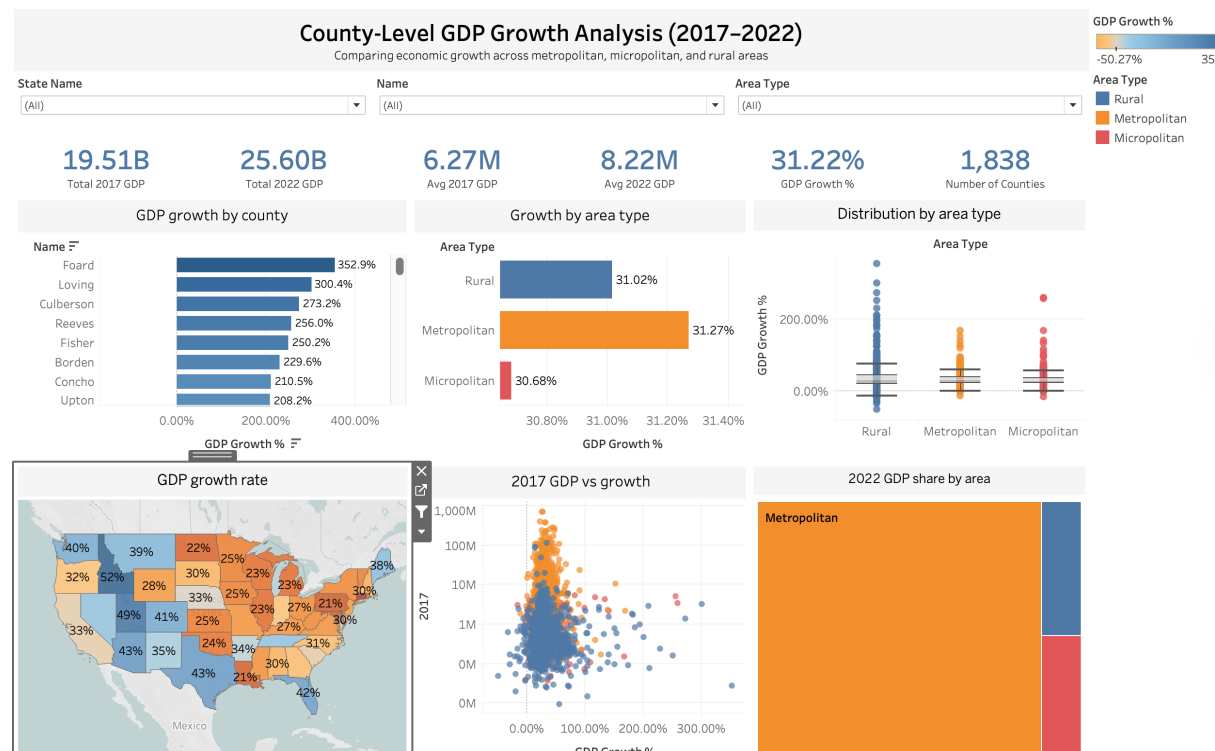


Figure 1: Overview of Tableau dashboard visualizing county-level population, employment, and housing price changes (2017–2022).

4 Results and Analysis

4.1 Summary Metrics

Aggregate metrics indicate steady national growth:

- Average Population Change: **+3.6%**

- Average Employment Change: **+4.8%**
- Average Housing Price Change: **+26.5%**

These results highlight robust economic recovery and expansion between 2017 and 2022.

4.2 Population Change Visualization

The bar charts and histograms show that most counties experienced positive population growth, with metropolitan areas leading the increase. Rural areas displayed more variability, including some counties with slight population decline, particularly in the Midwest.

4.3 Employment Change Visualization

Scatter plots comparing employment and population change rates show a strong positive relationship—counties with higher population growth tended to have stronger employment expansion. Metropolitan regions clustered near higher growth percentages, while rural regions exhibited wider distribution.

4.4 Housing Price Trends

The housing price change map revealed a notable surge across nearly all counties, with the highest growth concentrated in Western and Southern states. Rural regions saw larger relative increases, reflecting market adjustment after slower pre-2017 valuations.

4.5 Regional Comparison

Boxplots comparing the three classifications indicate:

- Metropolitan counties had the highest median values for population and employment.
- Rural counties had greater variance in housing price change.
- Micropolitan regions performed moderately in all categories.

5 Conclusion

This analysis demonstrates that economic growth between 2017 and 2022 was broadly distributed across the United States, though the composition of growth varied by region. Metropolitan counties continued to expand steadily, while rural areas experienced

more volatility but stronger relative housing growth. Tableau effectively captures these multidimensional trends, providing a dynamic overview of U.S. county-level economic change.

Appendix: R Script

```
library(tidycensus)
library(tidyverse)
library(plotly)
library(tigris)
options(tigris_use_cache = TRUE)

# Population
totalpop2022 = get_acs(geography="county", variables="B01003_001"
  ↪ , year=2022, geometry=TRUE)
totalpop2022 = totalpop2022 |> rename(totalpop22="estimate")

totalpop2017 = get_acs(geography="county", variables="B01003_001"
  ↪ , year=2017)
totalpop2017 = totalpop2017 |> rename(totalpop17="estimate")

popchange = full_join(totalpop2017, totalpop2022, by="GEOID") |>
  mutate(popchangerate2017_2022 = ((totalpop22 - totalpop17) /
    ↪ totalpop17) * 100) |>
  select(GEOID, NAME.x, totalpop17, totalpop22, popchangerate2017
    ↪ _2022, geometry) |>
  rename(NAME="NAME.x")

# Employment
totalemp2022 = get_acs(geography="county", variables="B23025_001"
  ↪ , year=2022) |> rename(totalemp22="estimate")
totalemp2017 = get_acs(geography="county", variables="B23025_001"
  ↪ , year=2017) |> rename(totalemp17="estimate")

empchange = full_join(totalemp2017, totalemp2022, by="GEOID") |>
  mutate(empchangerate2017_2022 = ((totalemp22 - totalemp17) /
    ↪ totalemp17) * 100) |>
  select(GEOID, NAME.x, totalemp17, totalemp22, empchangerate2017
    ↪ _2022) |>
  rename(NAME="NAME.x")

# Housing Price
totalhp2022 = get_acs(geography="county", variables="B25077_001",
  ↪ year=2022) |> rename(totalhp22="estimate")
```

```

totalhp2017 = get_acs(geography="county", variables="B25077_001",
  ↪ year=2017) |> rename(totalhp17="estimate")

houseprice = full_join(totalhp2017, totalhp2022, by="GEOID") |>
  mutate(housepricerate2017_2022 = ((totalhp22 - totalhp17) /
    ↪ totalhp17) * 100) |>
  select(GEOID, NAME.x, totalhp17, totalhp22, housepricerate2017_
    ↪ 2022) |>
  rename(NAME="NAME.x")

# Combine all datasets
dt = read.csv("classification.csv")
dt = dt |> select(GEOID, Metropolitan.Micropolitan.Statistical.
  ↪ Area)

data = full_join(popchange, empchange, by="GEOID") |>
  full_join(houseprice, by="GEOID")

data$GEOID = as.numeric(data$GEOID)
data = full_join(data, dt, by="GEOID") |>
  select(!(NAME.y)) |> rename(NAME="NAME.x")

data$Metropolitan.Micropolitan.Statistical.Area =
  data$Metropolitan.Micropolitan.Statistical.Area |> replace_na("
    ↪ Rural")

# Final data filter
finaldata1 = data |>
  filter(GEOID < 72001 | GEOID > 72153) |>
  filter(GEOID < 15001 | GEOID > 15009)

# Visualization
ggplot(finaldata1, aes(x=totalpop22, y=Metropolitan.Micropolitan.
  ↪ Statistical.Area)) +
  geom_boxplot(coef=1.5) + xlim(0, .05e+06)

```

References

- U.S. Census Bureau. “American Community Survey (ACS) Data.” <https://www.census.gov/programs-surveys/acs>
- U.S. Department of Commerce. “BEA Regional Economic Accounts.” <https://apps.bea.gov/regional/>
- Tigris and Tidycensus R Packages. “R Interface for Census and TIGER/Line Shapefiles.” <https://cran.r-project.org/web/packages/tidycensus/index.html>